PECH GROUP HOLDINGS LTD

AI Hardware & Setup Guide — Procurement, Installation, Configuration

Lagos, Nigeria — pechgroupholdings.tech

Document Reference: PECH-AI-HW-2026-001 **Version:** 1.0 **Effective Date:** March 2026 **Prepared By:** Technology Department **Approved By:** Managing Director / CEO

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1. HARDWARE STRATEGY OVERVIEW

Zero-Waste Philosophy

PECH's AI hardware strategy follows a **zero-waste single-platform** approach: buy the right platform from Day 1, add components as you grow, never throw anything away.

Phase	Timeline	Hardware	AI Capability
Phase 1 (Crawl)	Months 1-6	TRX50 Workstation + 1× RTX 4090	Run 2-3 models simultaneously; develop & test
Phase 2 (Walk)	Months 7-12	Same box + 2nd RTX 4090 + more RAM/storage	Production AI with dual-GPU; cloud burst for overflow
Phase 3 (Run)	Months 13-24	Co-locate same box + optional 2nd node	Full production; fine-tuned models; high concurrency

Every Phase 1 component carries through to Phase 3. Zero waste. Zero rebuilds. Zero downtime for migration.

Why Zero-Waste Matters

The old approach (buy a cheap AM5 starter, then replace with a TRX50 production box) wastes money:

Approach	Phase 1 Cost	Phase 2-3 Cost	24-Month Total	Waste
Old (Starter → Replace)	₦5.7M-₦7.7M	₦11M-₦15M (new box)	₦16.7M-₦22.7M	₦1.4M-₦1.9M (entire Phase 1 CPU, mobo, RAM, PSU, case)
Zero-Waste (TRX50 → Upgrade)	₦8.2M-₦11M	₦4.1M-₦5.3M (add GPU+RAM)	₦12.3M-₦16.3M	₦0
Savings	—	—	**₦4.3M-₦6.4M saved**	**100% reuse**

Key Decision: Own vs Cloud

Factor	Own Hardware	Cloud GPU

Monthly cost	₦0 after purchase (power only)	₦200K-₦2M/month
Upfront cost	₦8.2M-₦16.3M (Phase 1-2)	₦0
Latency	<10ms (local)	50-200ms (internet)
Data sovereignty	Full control (NDPA compliant)	Depends on provider
Scaling	Limited by hardware	Instant scale
Maintenance	PECH team responsibility	Provider handles
Power dependency	Lagos grid + backup	Provider's problem

PECH Recommendation: Start with **TRX50 Workstation + cloud GPU for overflow** (Phase 1). Add 2nd GPU when revenue justifies (Phase 2). Co-locate for production reliability (Phase 3).

2. PHASE 1 — TRX50 WORKSTATION (1 GPU)

Purpose: Run 2-3 AI models simultaneously for development, testing, and initial production. Built on a platform that scales to Phase 3 without replacing anything.

Bill of Materials (BOM)

#	Component	Specification	China Price (USD)	US Retail (USD)
1	**GPU**	NVIDIA RTX 4090 24GB GDDR6X	$1,200-$1,400	$1,600-$1,800
2	**CPU**	AMD Threadripper 7960X (24C/48T, sTR5)	$830-$1,100	$1,100-$1,400
3	**Motherboard**	ASUS Pro WS TRX50-SAGE WiFi (3× PCIe 5.0 x16, 8 DIMM)	$550-$700	$750-$950
4	**RAM**	128GB DDR5-5600 ECC RDIMM (4×32GB)	$280-$400	$400-$550
5	**Storage (OS)**	1TB Samsung 990 Pro NVMe Gen4	$50-$70	$80-$100
6	**Storage (Data)**	4TB Samsung 990 Pro NVMe Gen4	$200-$280	$300-$350
7	**PSU**	Seasonic PRIME TX-1600 80+ Titanium (ATX 3.1)	$350-$450	$450-$550
8	**Case**	4U Rackmount chassis (Rosewill RSV-L4500U)	$120-$180	$180-$280
9	**CPU Cooler**	Noctua NH-U14S TR5-SP6	$80-$110	$100-$130
10	**Fans**	3× 140mm Noctua NF-A14 case fans	$30-$45	$45-$60
	Subtotal		**$3,690-$4,735**	**$5,005-$6,170**

Why TRX50 from Day 1 (Not AM5)

Component	Why This Spec	Why Not AM5 (Cheaper)?
Threadripper 7960X	24 cores handle AI inference + web services + data pipelines. 128 PCIe 5.0 lanes for multi-GPU bandwidth	Ryzen 9 7950X has only 28 PCIe lanes — can't feed 2 GPUs properly. Entire CPU+mobo wasted at Phase 2
TRX50-SAGE	3× PCIe 5.0 x16 slots (2 for GPUs + 1 spare), 8 DIMM slots (up to 512GB), 4× M.2 NVMe	X670E only has 1× PCIe x16 + limited DIMM slots. No upgrade path
128GB DDR5 ECC RDIMM	Required by Threadripper (non-ECC won't work). ECC = production-grade reliability. 4 of 8 DIMM slots used — add 128GB more in Phase 2	AM5 uses UDIMM (consumer-grade, 4 slots max). If you start AM5, you throw away all RAM when moving to TRX50
1600W PSU	Sized for 2× RTX 4090 from Day 1 (dual-GPU peak ~1,200W). Zero power upgrade needed at Phase 2	A 1000W PSU can't handle 2 GPUs. You'd buy it, then replace it — wasted ₦150K-₦200K
4U Rackmount	Co-location ready from Day 1. Standard 19" rack compatible. Fits 2× full-length GPUs	A full-tower case can't go into a data center rack. You'd replace the case at Phase 3
Noctua TR5-SP6	Purpose-built for Threadripper sTR5 socket. Silent, 250W TDP rated	Different cooler mount from AM5 — another wasted component

The TRX50 premium is ₦2.5M-₦3.3M more than AM5 upfront. It saves ₦4.3M-₦6.4M by eliminating a full rebuild at Phase

2.

Quality Recommendations (Specific Brands)

Component	Recommended	Why This One
PSU	Seasonic PRIME TX-1600 (NOT generic 1600W)	80+ Titanium = 94% efficiency, 12-year warranty, ATX 3.1 native GPU power connector. No adapter cables needed for RTX 4090
Cooler	Noctua NH-U14S TR5-SP6 (NOT AIO liquid)	Air coolers have zero pump failure risk. Critical in Lagos heat (28-35°C ambient). Noctua has 6-year warranty
Motherboard	ASUS Pro WS TRX50-SAGE (NOT consumer boards)	Workstation-class VRM (16+2 phase), 10GbE LAN, IPMI remote management, validated for 24/7 operation
RAM	Samsung or SK Hynix ECC RDIMM (NOT third-party)	Threadripper is picky about RAM compatibility. Samsung/SK Hynix have the highest compatibility rates with TRX50
Case	Rosewill RSV-L4500U or Chenbro RM41300	Hot-swap bays, dual 120mm rear fans, tool-less rails. Standard co-lo form factor

What This Can Run (Simultaneously)

Slot	Model	VRAM Used	Purpose
1	Qwen2.5-7B (Q4)	4.5 GB	General AI assistant
2	Qwen2.5-Coder-7B (Q4)	4.5 GB	Developer platform
3	Qwen2.5-VL-3B (Q4)	2 GB	Image understanding
4	Faster-Whisper (Small)	1 GB	Voice input
5	RT-DETR-L	2 GB	Solar panel inspection
6	all-MiniLM-L6-v2	0.1 GB	RAG embeddings
	Total VRAM	**~14 GB**	**10 GB headroom on 24GB GPU**
CPU	Piper TTS, PaddleOCR, DistilBERT, XGBoost, Prophet	RAM only	CPU-based inference (24 cores handle it easily)

3. PHASE 2 — UPGRADE (ADD GPU + RAM + STORAGE)

Purpose: Full production deployment with dual GPUs, higher quality models, and concurrency. **This is NOT a new build — it's components added to the Phase 1 TRX50 Workstation.**

Upgrade BOM (Additions Only)

#	Component	Specification	China Price (USD)	US Retail (USD)
1	**GPU 2**	NVIDIA RTX 4090 24GB (or RTX 5090 if available)	$1,200-$1,400	$1,600-$1,800
2	**RAM**	+128GB DDR5-5600 ECC RDIMM (4×32GB, fills remaining 4 DIMM slots)	$280-$400	$400-$550
3	**Storage**	+4TB Samsung 990 Pro NVMe Gen4 (fills 3rd M.2 slot)	$200-$280	$300-$350
	Upgrade Subtotal		**$1,680-$2,080**	**$2,300-$2,700**

Component Reuse Matrix

Every Phase 1 component carries forward. Nothing is replaced:

Component	Phase 1	Phase 2	Phase 3	Status
CPU (Threadripper 7960X)	✓ Installed	✓ Kept	✓ Kept (co-located)	**Zero waste**
Motherboard (TRX50-SAGE)	✓ Installed	✓ Kept — 2nd GPU slot now used	✓ Kept	**Zero waste**
RAM (128GB ECC)	✓ 4 of 8 DIMMs	✓ All 8 DIMMs = 256GB	✓ Kept	**Zero waste**
PSU (Seasonic TX-1600)	✓ Already sized for 2 GPUs	✓ No change needed	✓ Kept	**Zero waste**
Case (4U Rackmount)	✓ Already co-lo ready	✓ No change needed	✓ Slides into rack	**Zero waste**

Cooler (Noctua TR5-SP6)	✓ Installed	✓ No change needed	✓ Kept	**Zero waste**
GPU 1 (RTX 4090)	✓ Slot 1	✓ Slot 1	✓ Slot 1	**Zero waste**
GPU 2 (RTX 4090)	—	✓ **NEW** — Slot 2	✓ Slot 2	Added in Phase 2
Storage (5TB)	✓ 2 of 4 M.2	✓ 3 of 4 M.2 = 9TB	✓ Kept	**Zero waste**

What This Can Run (Dual GPU)

GPU	Model	VRAM	Purpose
GPU 1	Qwen2.5-14B (Q4)	8 GB	Production AI assistant
GPU 1	Qwen2.5-Coder-14B (Q4)	8 GB	Full coding assistant
GPU 1	Faster-Whisper (Large-v3)	4 GB	Best accuracy STT
GPU 2	Qwen2.5-32B (Q4)	18 GB	Complex reasoning / premium tier
GPU 2	Qwen2.5-VL-7B (Q4)	5 GB	Production vision
CPU	All CPU-based models	RAM (256GB)	OCR, TTS, NLP, ML

4. PHASE 3 — CO-LOCATION & SCALING

Move the same TRX50 Workstation to a professional data center for production-grade reliability.

What Happens in Phase 3

Action	Details	Cost
Move TRX50 Workstation to co-lo	MainOne MDXi, Rack Centre, or 21st Century (Lagos)	₦100K-₦250K/month (rack + power + bandwidth)
4U rackmount slides directly into rack	No case change — already co-lo ready from Phase 1	₦0
Gain 99.99% uptime	Redundant power, precision cooling, 10Gbps+	Included in co-lo fee
Optional: Add 2nd node	Cheaper Ryzen 9 / used Threadripper for non-GPU work (databases, app servers)	₦3M-₦5M if needed

Monthly Cost After Co-Location

Component	Source	Monthly Cost
TRX50 Workstation (own)	Co-located at data center	₦0 (after purchase)
Co-location fee	MainOne / Rack Centre	₦100,000-₦250,000/month
Cloud GPU burst	RunPod / Vast.ai / Lambda	₦200K-₦500K/month
Total monthly		**₦300K-₦750K/month**

Note: Office power + cooling costs (₦150K-₦250K/month) are **eliminated** when you co-locate — the data center handles power and cooling. Net monthly increase is modest.

When to Use Cloud vs Local

Scenario	Use Local (TRX50)	Use Cloud
Development & testing	✓	
Low-traffic production (<50 concurrent)	✓	
High-traffic peaks (>50 concurrent)		✓
Large model inference (72B+)		✓
Fine-tuning / training		✓
Sensitive data processing	✓	
Batch processing (reports, OCR)	✓	

5. CHINA SOURCING GUIDE

Recommended Platforms

Platform	Best For	Language	Payment
1688.com (阿里巴巴)	Wholesale, bulk orders	Chinese (use agent)	Alipay, bank transfer
Taobao	Individual components	Chinese	Alipay
AliExpress	Small orders, international	English	Card, PayPal
JD.com (京东)	Genuine branded parts	Chinese	Card (international)
Shenzhen markets (in-person)	Best prices, inspect before buy	Chinese	Cash, bank transfer

Sourcing Strategy

Option 1: Use a Purchasing Agent (Recommended) - Companies like Superbuy, CSSBuy, or Nigerian-Chinese traders - Agent fee: 5-10% of order value - They handle: purchasing, quality inspection, consolidation, shipping - Best for: first-time buyers, quality assurance

Option 2: Direct from 1688/Taobao - Requires Chinese payment method (Alipay) - Use Shenzhen warehouse for consolidation - Ship via sea freight (cheapest) or air (fastest)

Option 3: Shenzhen Trip (Best Prices) - Visit Huaqiangbei electronics market - Inspect components in person - Negotiate bulk pricing - Combine with factory visits - Budget: $2,000-$3,000 for trip (flights, hotel, visa)

Supplier Verification Checklist

- [ ] Check seller rating (>4.8/5 on 1688)
- [ ] Request component photos with serial numbers
- [ ] Verify warranty terms (international warranty?)
- [ ] Ask for anti-static packaging guarantee
- [ ] Confirm GPU is genuine (check with GPU-Z upon receipt)
- [ ] For bulk: request factory visit or video call

Shipping from China to Lagos

Method	Cost (per server ~25kg)	Timeline	Best For
Sea freight	$150-$300	35-45 days	Non-urgent, bulk
Air freight	$400-$800	5-10 days	Urgent, single units
Express (DHL/FedEx)	$600-$1,200	3-5 days	Very urgent, small items

PECH Recommendation: Sea freight for bulk hardware. Air freight for GPU (high value, fragile).

Packaging Requirements

- **GPU:** Original box + anti-static bag + bubble wrap + "FRAGILE" labels
- **RAM:** Anti-static bag + foam padding
- **NVMe:** Anti-static bag (small, can ship with GPU)
- **PSU/Case:** Standard packaging (durable items)
- **All:** Double-box with 5cm foam gap between inner and outer box

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6. NIGERIA IMPORT PROCESS

Duties & Taxes

Tax	Rate	Applied To
CET (Common External Tariff)	5-20%	CIF value (varies by HS code)
VAT	7.5%	CIF + CET
CISS (Comprehensive Import Supervision Scheme)	1%	FOB value

ETLS levy	0.5%	CIF value
Surcharge	7%	CET value
Total estimated	**~25-35%**	On CIF value

HS Codes for Computer Parts

Component	HS Code	CET Rate
GPU (graphics card)	8471.80.00	10%
CPU (processor)	8542.31.00	5%
Motherboard	8473.30.00	10%
RAM / Memory	8542.32.00	5%
Hard drives / SSD	8471.70.00	10%
PSU / Cases	8504.40.00	20%
Complete computer	8471.49.00	20%

> **Pro Tip:** Import as individual components (not assembled computer) to get lower duty rates on most parts. HS code 8471.49 (complete computer) attracts 20% CET vs 5-10% for individual parts.

Clearing Agent

- Use a licensed customs clearing agent at Apapa/Tin Can ports (for sea) or MMIA (for air)
- Agent fee: ₦50,000-₦150,000 per shipment
- Documents needed: commercial invoice, packing list, bill of lading/airway bill, Form M, PAAR
- Timeline: 3-7 working days for customs clearance (Lagos)
- **Budget ₦100K-₦200K for clearing per shipment**

Import Checklist

- [ ] Get Form M from bank (required for imports >$50,000)
- [ ] Obtain PAAR (Pre-Arrival Assessment Report) from customs
- [ ] Ensure SONCAP certification (if applicable)
- [ ] Prepare commercial invoice with itemized components
- [ ] Insurance: 1-2% of CIF value (recommended for electronics)
- [ ] Assign clearing agent before goods arrive
- [ ] Budget for demurrage if clearing takes >7 days

7. LANDED COST CALCULATIONS

Phase 1: TRX50 Workstation — Landed Lagos

Cost Component	Amount (USD)	Amount (₦)
Hardware (China price)	$3,690-$4,735	₦5,535,000-₦7,102,500
Shipping (air freight, ~30kg)	$500-$900	₦750,000-₦1,350,000
Insurance (2%)	$74-$95	₦111,000-₦142,500
CIF Value	**$4,264-$5,730**	**₦6,396,000-₦8,595,000**
Import duties (~25%)	$1,066-$1,432	₦1,599,000-₦2,148,750
Clearing agent	$100	₦150,000
Local transport	$30	₦45,000
Total Landed	**$5,460-$7,292**	**₦8,190,000-₦10,938,750**

Round estimate: ₦8.2M-₦11M for Phase 1 TRX50 Workstation landed Lagos

Phase 2: Upgrade Components — Landed Lagos

Cost Component	Amount (USD)	Amount (₦)

Hardware (China price)	$1,680-$2,080	₦2,520,000-₦3,120,000
Shipping (air freight, ~5kg — GPU + RAM + SSD)	$200-$400	₦300,000-₦600,000
Insurance (2%)	$34-$42	₦51,000-₦63,000
CIF Value	**$1,914-$2,522**	**₦2,871,000-₦3,783,000**
Import duties (~25%)	$478-$630	₦717,750-₦945,750
Clearing agent	$80	₦120,000
Local transport	$20	₦30,000
Total Landed	**$2,492-$3,252**	**₦3,738,750-₦4,878,750**

Round estimate: ₦4.1M-₦5.3M for Phase 2 upgrade components landed Lagos

Zero-Waste vs Old Approach — 24-Month Hardware Comparison

	Old Approach	Zero-Waste	Difference
Phase 1 hardware	₦5.7M-₦7.7M (AM5 Starter)	₦8.2M-₦11M (TRX50)	+₦2.5M-₦3.3M
Phase 2 hardware	₦11M-₦15M (buy whole new TRX50 box)	₦4.1M-₦5.3M (add GPU+RAM only)	**-₦6.9M-₦9.7M**
Total hardware	**₦16.7M-₦22.7M**	**₦12.3M-₦16.3M**	**₦4.4M-₦6.4M saved**
Components wasted	CPU, mobo, RAM, PSU, case (~₦1.4M-₦1.9M)	None	**100% reuse**

> **Exchange rate used:** ₦1,500/$1 (March 2026 parallel market estimate). Adjust for current rates.

8. SUPPORTING INFRASTRUCTURE

Power Infrastructure That Scales (Sized for Phase 2 from Day 1)

In Lagos, Nigeria, hardware requires significant supporting infrastructure to operate reliably. **All infrastructure below is sized for 2× RTX 4090 (Phase 2) from Day 1** — so Phase 2 requires zero power infrastructure upgrades:

Power Infrastructure

Item	Specification	Cost (₦)	Why Essential
Online UPS	3kVA Pure Sine Wave (APC/CyberPower)	800,000-1,200,000	Protects against Lagos power fluctuations; provides 15-30 min battery backup
Solar Inverter System	5kVA Hybrid Inverter + 200Ah LiFePO4 batteries	2,000,000-3,500,000	Provides 4-8 hrs backup during NEPA outages; reduces diesel costs
Voltage Regulator	5kVA Stabilizer	80,000-150,000	Lagos voltage ranges from 160V-260V; servers need stable 220V
Diesel Generator	10kVA soundproof (backup to backup)	1,500,000-2,500,000	For extended outages >8 hrs; optional if solar is adequate
Dedicated Circuit	30A circuit from DB to server room	50,000-100,000	Isolate server power from office loads
Surge Protection	Industrial-grade SPD (Type 2)	30,000-60,000	Lightning protection (Lagos has severe thunderstorms)

Minimum power budget: ₦2.9M-₦5M Recommended power budget: ₦4.5M-₦7.5M (with solar + generator)

Power Consumption Estimate

Component	Idle (W)	AI Load (W)	Peak (W)
TRX50 Workstation — Phase 1 (1× RTX 4090)	180	600	750
TRX50 Workstation — Phase 2 (2× RTX 4090)	280	1,050	1,300
Networking (switch, router, firewall)	50	50	50
Cooling (AC unit)	500	1,000	1,500

UPS overhead (10%)	73	165	250
Total (Phase 1)	**803W**	**1,815W**	**2,550W**
Total (Phase 2)	**903W**	**2,265W**	**3,100W**

Monthly electricity cost (Phase 1, 24/7 AI load): ~1,310 kWh = ₦80,000-₦110,000 at Band A tariff **Monthly electricity cost (Phase 2, 24/7 AI load):** ~1,630 kWh = ₦100,000-₦135,000 at Band A tariff

> **Note:** The 3kVA UPS handles Phase 1 comfortably (1,815W peak < 2,400W capacity at 0.8 PF). For Phase 2 peak loads (2,265W), the UPS still covers AI workloads — only synthetic full-GPU torture tests would exceed capacity. The Seasonic TX-1600 PSU handles dual GPUs natively with no adapter cables needed.

Cooling Infrastructure

Item	Specification	Cost (₦)
Split AC Unit	2HP / 18,000 BTU (server room)	350,000-500,000
Backup AC Unit	1.5HP / 12,000 BTU (redundancy)	250,000-350,000
Temperature Monitor	IoT sensor with alert (ThingsBoard integrated)	15,000-30,000
Humidity Control	Dehumidifier (Lagos is 70-90% humidity)	50,000-100,000

Lagos ambient: 28-35°C. Server room target: 18-24°C, 40-60% humidity.

Cooling budget: ₦650K-₦1M

Network Infrastructure

Item	Specification	Cost (₦/month)
Primary Internet	Dedicated fiber 100Mbps (MainOne/Glo/MTN)	100,000-200,000
Backup Internet	4G/5G router (MTN/Airtel)	30,000-50,000
Managed Switch	24-port Gigabit PoE (TP-Link/Cisco)	100,000-250,000 (one-time)
Firewall	pfSense on Mini PC or Mikrotik RB4011	80,000-200,000 (one-time)
Ethernet Cabling	Cat6A (server room + office)	50,000-100,000 (one-time)
Static IP	Public static IP for remote access	Included or ₦10,000/mo

Network one-time: ₦230K-₦550K Network monthly: ₦130K-₦250K

9. SERVER ROOM SETUP

Minimum Requirements

Requirement	Specification
Size	Minimum 3m × 3m (9 sqm) dedicated room
Floor	Raised floor or anti-static tiles preferred
Door	Solid door with lock (access control)
Walls	No windows preferred (reduce heat ingress)
Power outlets	4× 15A sockets on dedicated circuit
Lighting	LED overhead (low heat)
Fire	CO2 or clean agent fire extinguisher (NOT water)

Server Rack Layout

```
 ┌──────────────────────────┐
│        SERVER ROOM (3×3m)  │
│                            │
│  ┌──────────┐   ┌──────────┐  │
```

```
│  │  12U Rack │      │   AC #1  │  │
│  │           │      │ (2HP)    │  │
│  │ [UPS    ] │      └──────────┘  │
│  │ [Switch ] │                    │
│  │ [Firewall]│      ┌──────────┐  │
│  │ [Server 1]│      │   AC #2  │  │
│  │ [Server 2]│      │ (1.5HP)  │  │
│  │ [Patch   ]│      └──────────┘  │
│  │ [Panel   ]│                    │
│  └──────────┘      ┌──────────┐  │
│                    │ Temp/    │  │
│  ┌──────────┐      │ Humidity │  │
│  │ Solar    │      │ Monitor  │  │
│  │ Inverter │      └──────────┘  │
│  │ + Battery│                    │
│  └──────────┘                    │
└──────────────────────────────────┘
```

Environmental Monitoring

Integrate with **ThingsBoard** (already in PECH IoT stack): - Temperature sensor (DS18B20 or DHT22) - Humidity sensor (DHT22) - Power monitoring (PZEM-004T) - Door sensor (magnetic reed switch) - Smoke detector (MQ-2 gas sensor) - ESP32 gateway → MQTT → ThingsBoard → Alerts

Cost: ₦30,000-₦60,000 for all sensors + ESP32

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10. SOFTWARE INSTALLATION GUIDE

Step-by-Step: Bare Metal → Production AI Server

Step 1: Operating System

```
# Install Ubuntu 22.04 LTS Server (NOT Desktop — saves resources)
# Download: https://releases.ubuntu.com/22.04/
# Flash to USB: use Rufus (Windows) or dd (Linux)

# After install:
sudo apt update && sudo apt upgrade -y
sudo apt install -y build-essential git curl wget htop nvtop
```

Step 2: NVIDIA Drivers + CUDA

```
# Add NVIDIA driver PPA
sudo add-apt-repository ppa:graphics-drivers/ppa -y
sudo apt update

# Install driver (check latest for RTX 4090)
sudo apt install -y nvidia-driver-545
sudo reboot

# Verify
nvidia-smi
# Should show RTX 4090, 24GB, driver version

# Install CUDA Toolkit 12.x
wget https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64/cuda-keyring_1.1-1_all.deb
sudo dpkg -i cuda-keyring_1.1-1_all.deb
sudo apt update
sudo apt install -y cuda-toolkit-12-4
```

Step 3: Docker + NVIDIA Container Toolkit

```
# Docker
curl -fsSL https://get.docker.com | sh
sudo usermod -aG docker $USER

# NVIDIA Container Toolkit (run AI models in Docker)
distribution=$(. /etc/os-release;echo $ID$VERSION_ID)
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-contair
curl -s -L https://nvidia.github.io/libnvidia-container/$distribution/libnvidia-container.list | \
  sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg] https://#g' | \
  sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list
sudo apt update && sudo apt install -y nvidia-container-toolkit
sudo nvidia-ctk runtime configure --runtime=docker
sudo systemctl restart docker

# Verify
docker run --rm --gpus all nvidia/cuda:12.4.0-base-ubuntu22.04 nvidia-smi
```

Step 4: Ollama (Phase 1 Model Management)

```
# Install Ollama
curl -fsSL https://ollama.com/install.sh | sh

# Pull Phase 1 models
ollama pull qwen2.5:7b          # General assistant (4.5GB)
ollama pull qwen2.5-coder:7b    # Code generation (4.5GB)
ollama pull nomic-embed-text    # Embeddings for RAG

# Test
ollama run qwen2.5:7b "Hello, I am PECH AI. How can I help?"
```

Step 5: Docker Compose — Full AI Stack

```
# docker-compose.yml — PECH AI Stack Phase 1
version: '3.8'

services:
  # Vector Database for RAG
  qdrant:
    image: qdrant/qdrant:latest
    ports:
      - "6333:6333"
    volumes:
      - qdrant_data:/qdrant/storage

  # PostgreSQL (primary database)
  postgres:
    image: postgres:16-alpine
    environment:
      POSTGRES_DB: pech
      POSTGRES_USER: pech_admin
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    ports:
      - "5432:5432"
    volumes:
      - pg_data:/var/lib/postgresql/data

  # Redis (cache)
  redis:
    image: redis:7.0-alpine  # Use 7.0 for BSD-3 license
    ports:
      - "6379:6379"
    volumes:
      - redis_data:/data

  # Faster-Whisper (Speech-to-Text)
  whisper:
    image: fedirz/faster-whisper-server:latest-cuda
    ports:
```

```
      - "8001:8000"
    deploy:
      resources:
        reservations:
          devices:
            - driver: nvidia
              count: 1
              capabilities: [gpu]
    environment:
      WHISPER__MODEL: small  # Use "large-v3" for production
      WHISPER__DEVICE: cuda

  # PaddleOCR (Invoice/Receipt OCR)
  paddleocr:
    image: paddlecloud/paddleocr:latest
    ports:
      - "8002:8866"

  # Monitoring
  prometheus:
    image: prom/prometheus:latest
    ports:
      - "9090:9090"
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml

  uptime-kuma:
    image: louislam/uptime-kuma:latest
    ports:
      - "3001:3001"
    volumes:
      - uptime_data:/app/data

volumes:
  qdrant_data:
  pg_data:
  redis_data:
  uptime_data:
```

Step 6: vLLM (Phase 2 — Production Serving)

```
 # When ready for production (higher throughput than Ollama)
pip install vllm

# Serve Qwen2.5-14B with OpenAI-compatible API
python -m vllm.entrypoints.openai.api_server \
  --model Qwen/Qwen2.5-14B-Instruct-AWQ \
  --quantization awq \
  --gpu-memory-utilization 0.9 \
  --max-model-len 8192 \
  --port 8000

# Now any OpenAI-compatible client can connect to http://localhost:8000
```

11. CLOUD GPU OPTIONS

For Phase 1-2 Overflow and Fine-Tuning

Provider	GPU Options	Price/hr (A100 80GB)	Billing	Nigeria Access
RunPod	A100, H100, RTX 4090	$1.89-$2.49	Per-second	Yes (crypto OK)
Vast.ai	Community GPUs	$0.80-$1.50	Per-second	Yes
Lambda Cloud	A100, H100	$1.99-$2.49	Per-hour	Yes
Google Colab Pro+	A100, V100	$49.99/month	Monthly	Yes
Paperspace	A100, RTX 4090	$1.89-$3.09	Per-hour	Yes

Recommended Strategy

Task	Provider	GPU	Cost Estimate
Fine-tuning (monthly)	RunPod	A100 80GB	$50-$200/session
Burst capacity	Vast.ai	RTX 4090	$0.40-$0.80/hr
Training experiments	Google Colab Pro+	A100	$50/month flat
Production overflow	RunPod Serverless	A100	Pay-per-inference

Monthly cloud budget estimate: ₦200,000-₦500,000

12. CO-LOCATION OPTIONS (LAGOS)

For Phase 3, consider co-locating production servers in a professional data center:

Data Center	Location	Price (per U/month)	Power	Connectivity
MainOne MDXi	Lekki, Lagos	₦50,000-₦100,000	Redundant	10Gbps+
Rack Centre	Oregun, Lagos	₦60,000-₦120,000	2N power	40Gbps+
21st Century	VI, Lagos	₦40,000-₦80,000	Redundant	1Gbps+
Galaxy Backbone	Multiple	₦45,000-₦90,000	Generator backup	10Gbps

Co-Location vs Office Server Room

Factor	Office Server Room	Co-Location
Power reliability	95% (with solar+gen)	99.99%
Cooling	2 ACs, manual monitoring	Precision cooling, 24/7
Security	Office security	Biometric, CCTV, guards
Internet	100Mbps fiber	1-10Gbps+
Monthly cost	₦150-₦250K (power+internet)	₦100-₦200K (co-lo) + existing power/internet
Setup cost	₦3.75-₦6.5M (infra)	₦100K-₦200K (install)

PECH Recommendation: - **Phase 1-2:** Office server room (lower total cost, faster setup) - **Phase 3:** Co-locate production servers at MainOne or Rack Centre (better uptime, cooling, connectivity)

13. MONITORING & MAINTENANCE

Daily Checks

```
 # GPU health
nvidia-smi

# Disk usage
df -h

# Docker containers
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"

# Ollama models loaded
ollama list
```

Automated Monitoring (integrate with PECH IoT stack)

Metric	Tool	Alert Threshold
GPU temperature	Prometheus + nvidia_exporter	>85°C
GPU memory usage	Prometheus	>90%
Disk usage	Prometheus + node_exporter	>85%

CPU load	Prometheus	>80% sustained
Room temperature	ThingsBoard (ESP32)	>28°C
Room humidity	ThingsBoard (ESP32)	>70%
UPS battery	SNMP / NUT	<50%
Internet uptime	Uptime Kuma	Any downtime
Model response time	Prometheus	>5s P95

Weekly Maintenance

- [ ] Check and clear Docker logs (`docker system prune`)
- [ ] Verify backup integrity
- [ ] Review GPU utilization trends
- [ ] Update Ollama models if new versions available
- [ ] Check disk space and archive old logs
- [ ] Verify UPS battery health
- [ ] Clean dust filters on server and AC

Monthly Maintenance

- [ ] Full system backup (rsync to external drive)
- [ ] Security updates (`sudo apt update && sudo apt upgrade`)
- [ ] Review cloud GPU spending
- [ ] AC filter cleaning
- [ ] Generator test run (15 minutes)
- [ ] Review model performance metrics

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14. TOTAL BUDGET SUMMARY

One-Time Costs (Zero-Waste Single-Platform)

Category	Phase 1 (₦)	Phase 2 Upgrade (₦)
Hardware (landed Lagos)	8,200,000-11,000,000	4,100,000-5,300,000
Power infrastructure	2,900,000-5,000,000	0 (already sized for 2 GPUs)
Cooling	650,000-1,000,000	0 (already sized)
Network (one-time)	230,000-550,000	0 (already set up)
Server room setup	200,000-400,000	0 (already set up)
IoT monitoring sensors	30,000-60,000	0 (already installed)
Total One-Time	**₦12,210,000-₦18,010,000**	**₦4,100,000-₦5,300,000**

Phase 1 midpoint: ~₦15.1M — comfortably within the ₦18M target cap.

Monthly Recurring Costs

Category	Monthly (₦)
Electricity	80,000-135,000
Internet (primary + backup)	130,000-250,000
Cloud GPU (overflow)	200,000-500,000
Maintenance/consumables	50,000-100,000
Total Monthly	**₦460,000-₦985,000**

24-Month Total Cost of Ownership

Setup Level	One-Time	Monthly × 24	24-Month Total
Phase 1 Only + Cloud	₦12.2M-₦18M	₦11M-₦23.6M	**₦23.2M-₦41.6M**
Phase 1 + Phase 2 + Cloud	₦16.3M-₦23.3M	₦11M-₦23.6M	**₦27.3M-₦46.9M**

How This Maps to PECH Financial Proposal (₦250M)

Proposal Line Item	Allocated (₦)	AI Hardware Fits?
Technology & Hardware	75,000,000	✓ Phase 1 hardware + infra = ₦12.2M-₦18M (16-24% of allocation)
Technology & Hardware	75,000,000	✓ Phase 1+2 hardware + infra = ₦16.3M-₦23.3M (22-31% of allocation)
Operational Expenses	28,800,000	✓ Monthly cloud + power + internet from here

Savings vs Old Approach

	Old (Starter → Replace)	Zero-Waste (TRX50 → Upgrade)	Saved
Phase 1	₦9.7M-₦14.7M	₦12.2M-₦18M	-₦2.5M (higher upfront)
Phase 2	₦11M-₦15M (whole new box)	₦4.1M-₦5.3M (add components)	**+₦6.9M-₦9.7M**
Net 24-month	₦20.7M-₦29.7M (one-time)	₦16.3M-₦23.3M (one-time)	**₦4.4M-₦6.4M saved**
Components wasted	~₦1.4M-₦1.9M	₦0	**100% reuse**

Verdict: The zero-waste TRX50 approach saves ₦4.4M-₦6.4M over 24 months while delivering better hardware from Day 1. AI hardware fits comfortably within the ₦75M technology allocation.

This document is confidential to PECH Group Holdings Ltd. Last updated: March 2026.